RiskPulse AI

From unstructured text to risk signals, index weights and stress tests

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Problem & Approach

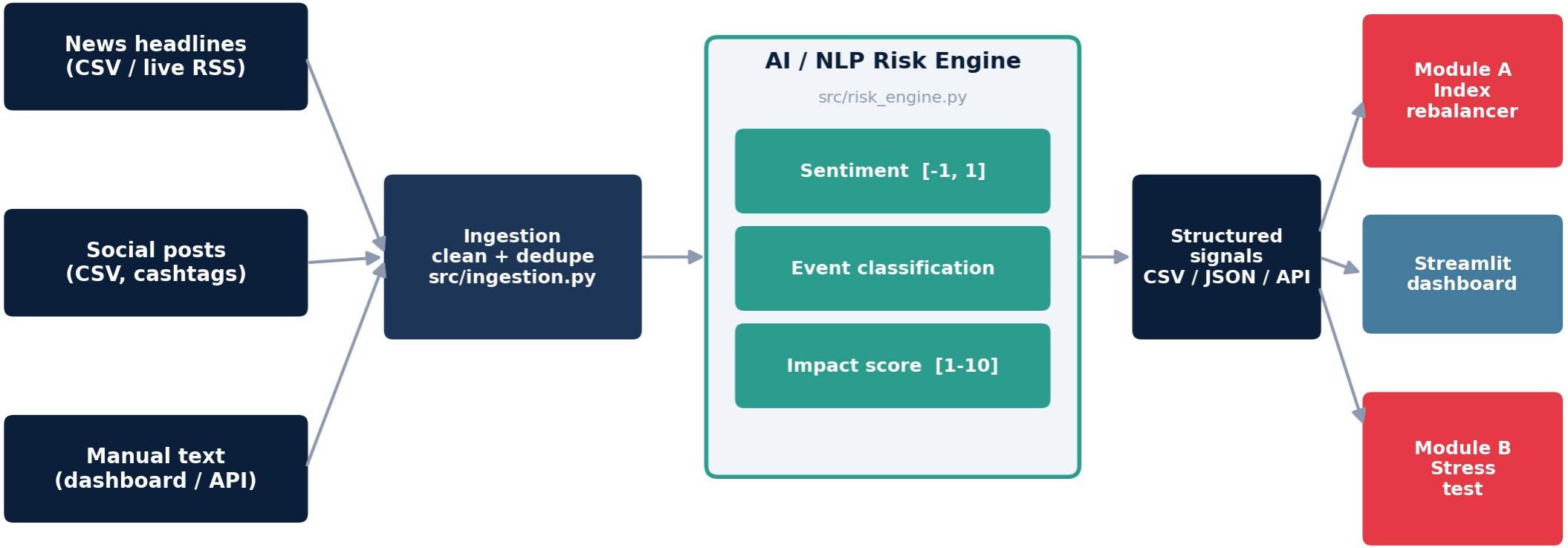
The problem

- Risk news arrives as free text, faster than anyone can read it
- Desks need structured, comparable signals, not headlines
- Those signals must feed real decisions, not just a score

What I built

- NLP engine: sentiment (-1 to 1), event type, impact (1-10)
- Two sources: news headlines and social posts
- Module A: sentiment-driven index rebalancer
- Module B: event-driven portfolio stress test

System Design



Sentiment drives Module A weights | Event type + impact drive Module B shocks

Both modules read the same signal table; the dashboard, API and CSV/JSON export serve it to users.

Implementation Highlights

Risk engine

Lexicon sentiment with negation and intensifiers, word-boundary event rules, explainable impact score. Every output lists the words behind it.

Index rebalancer

Per-stock sentiment state with half-life decay, exponential tilt, 2%-15% weight caps, turnover tracking.

Stress tester

Bonds and loans shocked via duration and spread duration; severity scales the shock; dashboard skips stress on positive news by default.

Delivery

Streamlit dashboard, FastAPI endpoints, CSV/JSON export, 24 automated tests, labelled evaluation set.

Key Results

84%

event-type accuracy (baseline 18%)

60%

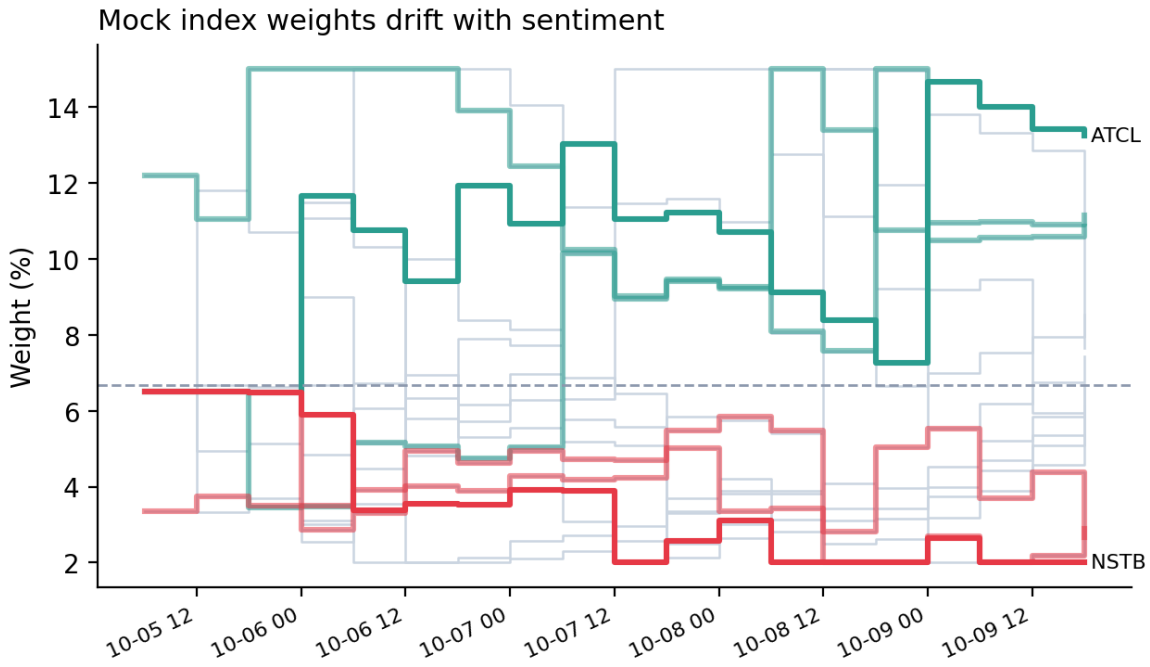
sentiment accuracy (baseline 43%)

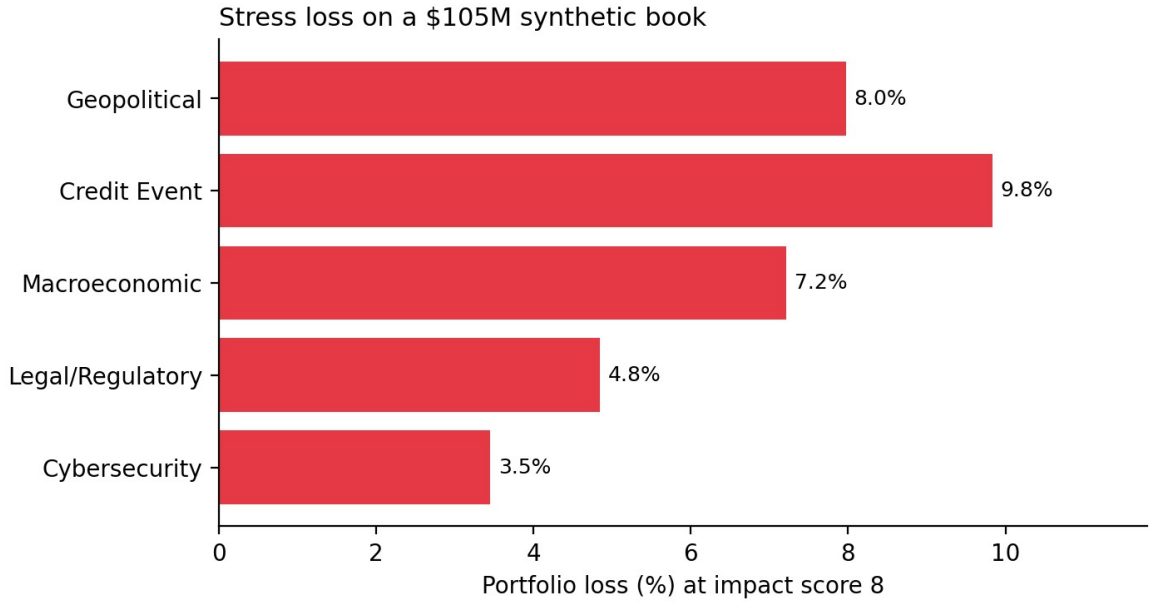
~10%

loss on the $105M book, credit event at impact 8

15

stocks rebalanced across 5 days of signals





Domain Impact

- Turns text into a traceable signal: every score shows the words that produced it
- One signal, two time horizons: fast tactical weights and slower strategic stress losses
- Shows how an event type maps to different damage by asset class (e.g. gov bonds can gain in a geopolitical shock)
- Transparent rules give risk teams something they can audit and challenge before trusting a model
- Modular design: swap the engine for a transformer without touching the downstream modules

Limitations & Next Steps

Limitations

- Sentiment is a lexicon baseline; misses phrases like "tensions ease"
- Impact is a heuristic, not calibrated to returns
- Synthetic data and a small hand-labelled test set
- Shocks ignore correlations, liquidity and Greeks

Next steps

- Fine-tune a finance transformer (FinBERT-style)
- Larger labelled set and historical backtests
- Calibrate impact against realised moves
- Streaming ingestion and richer stress models